

PROJECT REPORT

Car Rental Request Automation in ServiceNow Using Service Catalog and Workflows

Date: 23 September 2026 | Team Member: Vempali Sampath Kumar

1. INTRODUCTION

None

1.1 Project Overview

The Car Rental Request Automation project is implemented on the ServiceNow platform using a Service Catalog item and Flow Designer. It provides a structured way for a requester to submit car rental requirements and automates the subsequent approval and fulfillment process.

1.2 Purpose

The purpose is to reduce manual coordination by collecting request information in a standard form, routing the Requested Item for approval, creating a fulfillment Catalog Task after approval, and generating an email notification.

2. IDEATION PHASE

None

2.1 Problem Statement

Car rental requests can require repeated coordination between the requester, approver and fulfillment team. Without a structured process, approval status and fulfillment actions may be difficult to track.

2.2 Empathy Map Canvas

The requester wants quick processing, clear approval/status information and timely fulfillment. Pain points include manual coordination, waiting for approval and uncertainty about progress. Gains include a structured process, automated approval/fulfillment and timely notification.

2.3 Brainstorming

The main ideas identified were a Service Catalog request form, approval automation, Catalog Task creation, assignment to a fulfillment group, email notification and possible future business-rule enhancements. The implemented high-priority flow uses the request form, approval, task creation and email notification.

3. REQUIREMENT ANALYSIS

None

3.1 Customer Journey Map

The journey is: identify rental need → open Service Catalog → enter rental details → submit request → wait for approval → receive approval outcome → fulfillment task is created → notification is generated.

3.2 Solution Requirement

Functional requirements include capturing Requested Date, Pickup Location, Drop Location, Duration, Car Type and Reason; creating a Requested Item; requesting approval; creating a Catalog Task after approval; and generating an email notification. Non-functional requirements include traceability, usability, maintainability and controlled access within ServiceNow.

3.3 Data Flow Diagram

At a high level, the requester submits rental information through the Service Catalog. ServiceNow creates the Requested Item. Flow Designer reads the request and initiates approval. An approved request produces a Catalog Task for fulfillment and triggers the configured email action.

3.4 Technology Stack

Platform: ServiceNow. Request interface: Service Catalog. Automation: Flow Designer / Workflow Studio. Data records: Requested Item, Approval and Catalog Task. Notification: ServiceNow Send Email action. Development/testing environment: ServiceNow Developer Instance.

4. PROJECT DESIGN

None

4.1 Problem Solution Fit

The solution directly connects the identified coordination problem with a standardized ServiceNow request and workflow. It gives the requester a structured form and gives the fulfillment team an actionable task after approval.

4.2 Proposed Solution

Create the Car Rental Request catalog item, capture required rental details, attach the Car Rental Req flow, request approval, evaluate the approval condition, create a Catalog Task for the configured fulfillment group, and generate an approval email.

4.3 Solution Architecture

The architecture consists of the requester, Service Catalog, Requested Item, approval process, Catalog Task, fulfillment group and email notification. The successful execution followed: Ask For Approval → If Approved (True) → Create Catalog Task → Send Email.

5. PROJECT PLANNING & SCHEDULING

None

5.1 Project Planning

The project was organized into requirement/design work followed by Service Catalog configuration, workflow automation, testing and documentation. Story-point estimates were used to plan the work and a burndown representation was prepared.

6. FUNCTIONAL AND PERFORMANCE TESTING

None

6.1 Performance Testing

The recorded completed Flow Designer execution took approximately 2.266 seconds. Observed action timings were approximately 219 ms for Ask For Approval, 2042 ms for the approval branch, 476 ms for Create Catalog Task and 1462 ms for Send Email. These are observed test-run values, not general platform performance guarantees.

7. RESULTS

None

7.1 Output Screenshots

The completed project produced a Requested Item with an Approved state, a Catalog Task (SCTASK0010001), an email record with the subject "Your Car Rental Request is approved.", and a completed Flow Designer execution.

8. ADVANTAGES & DISADVANTAGES

None

8.1 Advantages

Structured request capture; automated approval; traceable Requested Item and Catalog Task; reduced manual coordination; clear workflow execution; email notification.

8.2 Disadvantages

The current implementation is a project-scale workflow and does not include advanced Performance Analytics dashboards, real-time vehicle inventory integration or complex multi-level approval. Outbound email delivery also depends on the instance email configuration.

9. CONCLUSION

The project demonstrates how ServiceNow can automate a car rental request process from request submission through approval, fulfillment task creation and notification. The tested workflow completed successfully and provides a traceable request-to-fulfillment process.

10. FUTURE SCOPE

Future enhancements can include Performance Analytics dashboards, vehicle availability integration, additional approval levels, richer reporting, automated reminders, SLA tracking and expanded notification channels.

11. APPENDIX

None

11.1 Source Code

No standalone source-code repository is required for the core ServiceNow configuration. The project consists primarily of ServiceNow configuration, catalog variables, Flow Designer automation and platform records.

11.2 Dataset Link

No external dataset is required for this workflow project.

11.3 GitHub & Project Demo Link

GitHub repository: <https://github.com/sampathvempali/car-rental-request-automation-servicenow> Project demo video: add the final demo-video URL here when available.